European Organisation for Astronomical Research in the Southern Hemisphere

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APPLICATION FOR OBSERVING TIME

PERIOD: **100A**

Important Notice:

By submitting this proposal, the PI takes full responsibility for the content of the proposal, in particular with regard to the names of CoIs and the agreement to act according to the ESO policy and regulations, should observing time be granted.

1. Title

Category: **C–7**

Understanding stellar activity in exoplanet-search data: the first joint high-precision photometry, radial velocity, and polarimetry campaign

2. Abstract / Total Time Requested

Total Amount of Time: 5.6 nights VM, 0 hours SM

The detection and characterisation of extrasolar planets is currently limited by the variability of the host star, mainly caused by magnetic activity (e.g. star spots, faculae, magnetic cycles). This will affect the scientific exploitation of next generation exoplanet-hunter instruments, such as SPIRou, ESPRESSO, and the TESS and PLATO space missions. To overcome this limitation, we propose to perform for the first time a simultaneous high-precision photometry, radial velocity, and polarimetry campaign on a small sample of 4 solar-type stars that already show activity signals in photometry and Zeeman signatures. This is a unique opportunity thanks to the particularity of one of the campaigns of the Kepler/K2 space telescope. With 5n of HARPS time, we aim at understanding in depth the impact of stellar activity in the precise radial velocity data. This is the required fundamental step to improve the methods used to correct for stellar activity in exoplanet-search data.

3.

Run	Period	Instrument	Time	Month	Moon	Seeing	Sky	Mode	Type
A	100	HARPS	5.6n	jan	g	n	THN	v	

4. Number of nights/hours

	Telescope(s)	Amount of time
a) already awarded to this project:		
b) still required to complete this project:		

5. Special remarks:

If granted time, we will share the telescope time with the Geneva pool of observers (LP PIs: Bonfils, Santerne) to increase the time coverage of the observations and decrease mission costs. This proposal is time critical and the observations should be performed simultaneously with the campaign C16 of the K2 space mission (dates: 2017/12/07 to 2018/02/25, secured to within a few days). This proposal is linked with other proposals requesting 47h of ESPaDOnS@CFHT (submitted), 12n of NARVAL@TBL and 15n of SOPHIE@OHP (deadline April 10), and 2n of CARMENES@CalarAlto (deadline April 3), to complement these HARPS data.

6. Principal Investigator:

Alexandre Santerne, alexandre.santerne@lam.fr, F, Laboratoire d'Astrophysique de Marseille (LAM)

6a. Co-investigators:

C.	Moutou	Canada-France-Hawaii Telescope Corporation,F
J.-F.	Donati	IRAP (Institut de Recherche en Astrophysique et Planetologie), UMR 5277,F
I.	Boisse	Laboratoire d'Astrophysique de Marseille (LAM),F
T.	Lopez	Laboratoire d'Astrophysique de Marseille (LAM),F

Following CoIs moved to the end of the document ...

7. Description of the proposed programme

A – Scientific Rationale: The study of extrasolar planets is unveiling an amazing diversity of exoplanets, from extremely hot Earth-size planets orbiting very close to their star (e.g. Kepler-78b; Pepe et al., 2013, Nature, 503, 377) up to giant planets orbiting at wide separation (e.g. HR8799; Marois et al., 2010, Nature, 468, 1080), through planets in the habitable zone (e.g. Proxima Centauri b; Anglada Escude et al., 2016, Nature, 536, 437 and TRAPPIST-1; Gillon et al., 2017, Nature, 542, 456). Those detections provide us with very important constraints to understand **the formation mechanisms of planetary systems** in general, including the solar system and the Earth. It is also a first step towards the search for life in the Universe and the conditions needed for it to develop.

The most important fundamental parameter of exoplanets is **their mass**, needed e.g. to distinguish between populations of planets (e.g. Mayor et al., 2011, arXiv:1109.2497) and to infer their internal structure (e.g. Brugger et al., 2016, ApJL, 831, L16), especially once combined with the radius measurement from a photometric transit. Exoplanet masses are mainly measured using **the radial velocity technique** (RV), consisting in observing the Doppler reflex motion of the star orbited by the planet. However, this technique is also very sensitive to the intrinsic **variability of the host star** which is able to mimic the presence of a planet (e.g. Queloz et al., 2001, A&A, 379, 279) or challenge its characterisation (see for instance the mass determination of the super-Earth CoRoT-7 b; e.g. Haywood et al., 2014, MNRAS, 443, 2517 among many other studies). This limitation is particularly critical in **the regime of low-mass planets** whose RV signal has a similar amplitude and periodicity as stellar active regions (such as star spots, faculae for instance). The Doppler signal of an Earth-like planet (9cm/s) is much lower than the signal caused by stellar variability (a few m/s for relatively quiet stars; Isaacson & Fischer, 2010, ApJ, 725, 875).

Stellar activity is currently the main astrophysical challenge to the exploration of planetary systems and will affect the scientific exploitation of the next-generation exoplanet-hunter instruments, both on the ground (e.g. the SPIRou@CFHT and ESPRESSO@VLT spectrographs) and in space (e.g. the NASA TESS and the ESA PLATO missions) as the latter rely on exoplanet mass measurements to reach their scientific objectives. Therefore, it is urgent to make substantial progress towards the understanding of stellar activity and how it affects RV measurements. **Therefore, in this proposal, we want to substantially improve our understanding of stellar activity to then better correct for it in the exoplanet-search data.**

The golden approach for this is to **combine techniques with dense monitoring of the stellar rotation period**. A few attempts have been done so far to combine simultaneous space-based photometry and precise ground-based radial velocities (in the cases only of the HD189733 and CoRoT-7 systems; Lanza et al., 2011, A&A, 533, A44; Haywood et al., 2014, respectively). Using the exquisite-precision photometric data of these two active K dwarfs, it was however not possible to fully correct the radial velocities. The main reason for this is the high **degeneracy in the modelling of the photometric data**: different combinations of dark spots and bright faculae can produce the same light curves (e.g. Bruno et al., 2016, A&A, 595, A89), whereas they have a different impact on the RV (see for instance the results for the most advanced tool to simulate activity feature on solar-type stars: SOAP 2.0; Dumusque, Boisse & Santos, 2014, ApJ, 796, 132).

The key missing ingredient of these studies was the characterisation of the **magnetic field of the star provided by spectro-polarimetry**. This was found to be the most efficient diagnosis to model radial-velocity variations caused by magnetic activity (Hébrard et al., 2014, MNRAS, 443, 2599, 2016, MNRAS, 461, 1465, Haywood et al., 2016, MNRAS, 457, 3637). Therefore, in this proposal, we want to conduct **for the first time, a simultaneous high-precision photometry, and ground-based RV, and spectro-polarimetry campaign on a sample of 4 relatively bright and active G- to M-dwarf stars**. This will be possible by combining the data from the NASA Kepler/K2 space mission, the HARPS@ESO, SOPHIE@OHP and CARMENES@CalarAlto RV spectrographs together with the ESPaDOnS@CFHT and NARVAL@TBL spectropolarimeters, over several stellar rotation periods.

This campaign is only possible because the Kepler telescope will observe during the K2 campaign C16 in forward-facing mode, i.e. pointing towards the Earth, which will allow large simultaneous ground- and space-based observations. This is not possible during normal K2 campaigns (in rear-facing mode). Over the 3 year lifetime of the K2 mission, only the campaigns C9 and C16 are done in the forward-facing mode (note that C9 was dedicated to the galactic center for microlensing detections). The K2 campaign C16 (in operation from 2017 Dec. 7 to 2018 Feb. 25) is therefore a unique opportunity to perform such simultaneous photometry, RV, and spectropolarimetry campaign. Moreover, the field-of-view of C16 will overlap the one of C5. As a consequence, more than 3000 stars that will be observed early 2018 were already observed by K2 mid-2015 (see Fig. 1), allowing us to select the best stars for this campaign (see box TargetNotes for details about the target selection).

B – Immediate Objective: We want to use HARPS to measure precise (at the level of 1-2 m/s) radial velocity variations of the stars over a few stellar rotation periods and simultaneously with the photometry and polarimetry campaign. For that, we estimated that we need 30 measurements for each stars with a cadence of 1 point per star every nights. This will allow us to cover about 1.5, 2, 3, or 4 rotation periods (depending on the star - see Table 1).

7. Description of the proposed programme and attachments

Description of the proposed programme (continued)

With this simultaneous photometry, RV, and polarimetry campaign, we aim at:
1. exploring how the radial velocity, line width and bisector as well as the activity indices (e.g. Ca II H&K, H_α, NaD, He I) change with respect to the photometry and the instantaneous magnetic field ;
2. deriving the magnetic field topology of the star and comparing it with the positions of the active regions as obtained by modelling the 3-month K2 light curve (as in Bruno et al., 2016).
3. testing the accuracy of stellar activity codes (such as SOAP 2.0) in the modelling of both the photometric and spectroscopic data, given the constraints from the stellar magnetic map. If these codes cannot reproduce the data, then we will explore what their missing physical ingredients are.
4. probing if differences in the behaviour of the activity features depends on the spectral type, in particular regarding the respective role of dark spots and bright faculae on the RV activity jitter.
As a consequence, with this photometry, RV and polarimetry simultaneous campaign, and for the first time, we will deeply study stellar active (magnetic) regions and their impact on data sets used to search and characterise small exoplanets. This will not only allow us to better understand the variability of stars, it will also substantially improve the methods to correct for this variability in the data. **This is a fundamental step to fully exploit the new generation instruments (e.g. SPIRou, ESPRESSO, TESS, PLATO) that aims at searching and characterising small worlds in the habitable zones of their host stars.**

Attachments (Figures)

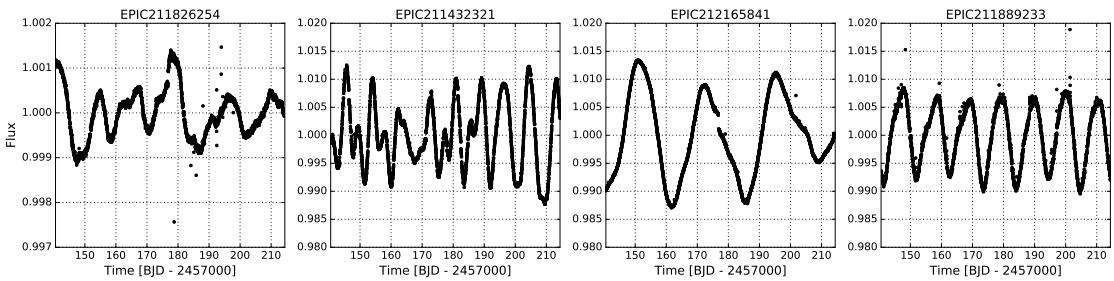

Fig. 1: K2 light curve of the 4 selected stars obtained mid-2015 during the campaign C5. Note the different flux scale between the first (left) panel and the three following (right) ones.

Target	Teff [K]	P_{rot} [d]	Phot. Var. [mmag]	V mag	exp. RV jitter [m/s]	Texp [s]	exp. σ_{RV} [m/s]
EPIC211826254	5375	14	1	9.3	3.5	900	1.7
EPIC211432321	4680	7.5	10	10.9	57	900	3.0
EPIC212165841	4530	21	10	10.8	20	1200	2.5
EPIC211889233	3860	10	7	12.2	21	1800	3.1

Table. 1: Main parameters of the 4 selected stars. The photometric variability is half the peak-to-peak variation. For comparison the active sun has a photometric variability (in a similar bandpass) at the level of a 1-3 mmag (Basri et al., 2013, ApJ, 769, 37). The last 3 columns are the expected RV semi-amplitude of the activity signal, the requested exposure time per observation and the associated ETC-estimated RV photon noise.

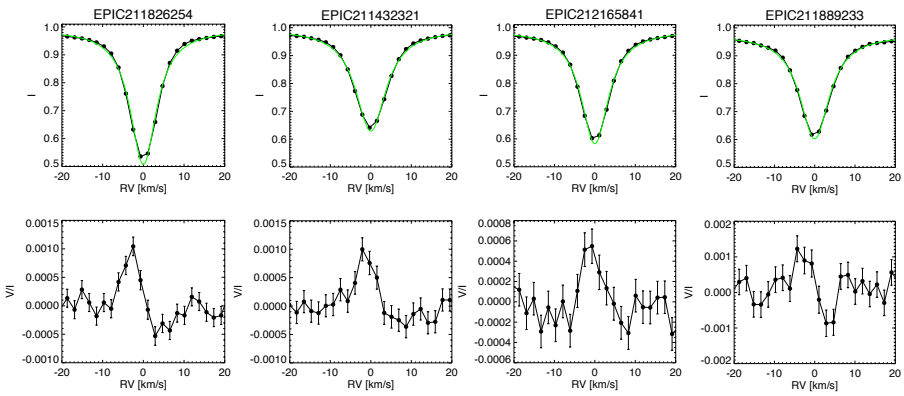

Fig. 2: Intensity and Stokes V profiles of the four targets obtained with ESPaDOnS in March 2017 (DDT), showing detections of the magnetic signatures.

8. Justification of requested observing time and observing conditions

Lunar Phase Justification: The targets are located close to the ecliptic plane. As consequence, the Moon might be very close to the targets. We prefer grey time to limit contamination in the spectra from the Moon light. However, this will adjusted with the other programmes sharing time with the Geneva pool.

Time Justification: (including seeing overhead) We request 30 observations over 30 nights on the four selected targets. Each night, we need (900+900+1200+1800) = 4800s + 600s (overheads) = 5400s = 1.5h. 1.5h x 30 epochs = 45 h = 5.6n.
Note that the visibility of the targets is not excellent from La Silla, but still they are all visible for at least 2 hours and up to 5 hours above airmass 1.7 over the K2 campaign C16 (December - February).

8a. Telescope Justification:

HARPS is nothing less than the best radial velocity machinery in the world. HARPS is therefore a critical instrument for this project in order to detect faint feature at the level of a few m/s. We did not request HARP-Spol for several reasons: 1. ESPaDOnS and NARVAL are more efficient spectrographs to measure polarimetric signals, 2. the visibility of the targets is better from these observatories, 3. it would have substantially increase the amount of time requested on the highly requested HARPS spectrograph. We request HARPS in objAB to monitor the background sky. We do not aim at RV precision below 1m/s, so the Thosimult or WAVE mode are not needed.
SOPHIE@OHP will complement the HARPS observations with a longer coverage but lower RV precision and CARMENES@CalarAlto will complement with NIR RV observations, a wavelength domain where the activity signal is expected to be lower.

8b. Observing Mode Justification (visitor or service):

Only visitor mode is available at La Silla

8c. Calibration Request:

Standard Calibration

9. Report on the use of ESO facilities during the last 2 years

Large Programmes on HARPS 198.C-0169 (PI: Santerne), 196.C-0042 (PI: Santos), 198.C-0836 (PI: Díaz) to search and characterise low-mass planets around solar-type stars. With only a few months of data, we are preparing several planet-discovery papers, including a planet of about 3 Mearth. However, the activity of stars is a limitation to either fully exploit our data or to reach an exquisite precision on the planet fundamental parameters. Therefore this timely programme will allow us to better exploit our large programme data.

9a. ESO Archive - Are the data requested by this proposal in the ESO Archive (http://archive.eso.org)? If so, explain the need for new data.

The requested time-critical observations are not already in the archive

9b. GTO/Public Survey Duplications:

10. Applicant's publications related to the subject of this application during the last 2 years

Boisse, I., Bouchy, F., Hebrard, G., et al. 2011, A&A, 528, A4: Disentangling between stellar activity and planetary signals

Boisse, I., Bonfils, X., & **Santos, N.** 2012, A&A, 545, A109: SOAP A tool for the fast computation of photometry and radial velocity induced by stellar spots

Bruno, G., **Deleuil, M.**, **Almenara, J.-M.**, et al. 2016, A&A, 595, A89: Disentangling planetary and stellar activity features in the CoRoT-2 light curve

Donati, J.-F., Hebrard, E., Hussain, G., et al. 2014, MNRAS, 444, 3220: Modelling the magnetic activity and filtering radial velocity curves of young Suns : the weak-line T Tauri star LkCa 4

Dumusque, X., **Boisse, I.**, & **Santos, N.** 2014, ApJ, 796, 132: SOAP 2.0: A Tool to Estimate the Photometric and Radial Velocity Variations Induced by Stellar Spots and Plages

Haywood, R. D. , Collier Cameron, A., Unruh, Y. C. et al., 2016, MNRAS, 457, 4: The Sun as a planet-host star: Proxies from SDO images for HARPS radial-velocity variations

Hebrard, E., **Donati, J.-F.**, Delfosse, X., et al. 2014, MNRAS, 443, 2599: Detecting planets around active stars: impact of magnetic fields on radial velocities and line bisectors

Hebrard, E, **Donati, J.-F.**, Delfosse, X., et al. 2016, MNRAS, 461, 1465: Modelling the RV jitter of early -M dwarfs using tomographic imaging

Santerne et al., in prep.: Precise planetary parameters from the ESO-K2 large programme. I. K2-XX b: a 3 Mearth rocky planet transiting a very active K dwarf

11. List of targets proposed in this programme

Run	Target/Field	α(J2000)	δ(J2000)	ToT	Mag.	Diam.	Additional info	Reference star
A	EPIC211826254	08 50 36.87	+17 41 21.6	7.5	9.2	0		
A	EPIC211432321	08 44 54.59	+12 07 48.5	7.5	10.9	0		
A	EPIC212165841	08 56 56.62	+23 23 20.4	9	10.8	0		
A	EPIC211889233	08 43 44.00	+18 34 07.2	20	12.2	0		

Target Notes: Target selection: Among the 3000+ stars overlapping the C5 and C16 of K2, we selected 3 G-, 4 K-, and 3 M-dwarfs as being the brightest of their spectral type. Their K2 light curve collected in 2015 exhibit clear activity signal ranging various rotation periods and photometric variability. We performed snapshot observations with ESPaDOnS in March 2017 (thanks to a DDT) on these 10 targets and detected Zeeman signatures for 5 of them. We finally rejected one of them because its rotation period was too long for a good monitoring in polarimetry (24d) and its active regions were evolving too quickly as revealed by the photometry. We end up with a sample of 4 relatively bright dwarf stars (1 G-, 2 K-, and 1 M-dwarf) with marginal to definite Zeeman signatures already detected with ESPaDOnS (see Fig. 2), with rotation periods ranging from 7.5 to 21 days, and photometric variability (PhotVar) ranging from 1 to about 10 mmag (see Table 1). Their C5 K2 light curves are shown in Fig. 1. None of these targets is known to host a multiple planetary or stellar system so far.

We estimated the RV semi-amplitude of the activity signal (see Table 1) using a straithforward relation based on the stellar radius, rotation period, and the photometric variability. This relation was validated on the few active stars well observed in RV (e.g. CoRoT-7, HD189733).

12. Scheduling requirements

This proposal involves time-critical observations, or observations to be performed at specific time intervals.

12. Scheduling requirements contd...

4. Specific date(s) for time critical observations:

Run	from	to	reason
A	07-dec-17	25-feb-18	Scheduled dates for the K2 C16 campaign.

13. Instrument configuration

Period	Instrument	Run ID	Parameter	Value or list
100	HARPS	A	spectro-ObjA(B)	HARPS

6b. Co-investigators:

...continued from Box 6a.

M.	Deleuil	Laboratoire d'Astrophysique de Marseille (LAM),F
M.	Hobson	Laboratoire d'Astrophysique de Marseille (LAM),F
N.C.	Santos	IA - Instituto de Astrofisica e Ciencias do Espaco,P
D.	Barrado Navascués	Centro de Astrobiologia (CSIC-INTA),E
J.	Lillo-Box	ESO Office Santiago,ESO
D.J.	Armstrong	University of Warwick,UK
J.M.	Almenara	Observatoire Astronomique de l'Universite de Geneve,CH
R. D.	Haywood	Harvard-Smithsonian Center for Astrophysics,US
S.	Barros	IA - Instituto de Astrofisica e Ciencias do Espaco,P
L.	Serrano	IA - Instituto de Astrofisica e Ciencias do Espaco,P
R.F.	Díaz	Instituto de Astronomia y Fisica del Espacio (IAFE),AR